Springer morphisms for F4 E6 and E7

Independent calculation for the minimal Dynkin index representations

Prepared for Shrawan Kumar 16 October 2026

This note applies the uniform weight-matrix formula of Sean Rogers [3, Theorems 1 and 2] to $V_{F_4}(\omega_4)$, $V_{E_6}(\omega_1)$ and $V_{E_7}(\omega_7)$. The independent verification consists of regenerating the weights from root data and reevaluating the matrices and coefficients, separately from the numerical calculations in [1] and [2]. The underlying mathematical method is Rogers's formula.

The F4 and E7 coordinate formulas agree with Lemmas 6.1 and 8.1 of [1]. For E6, all six denominators in Lemma 7.1 must change from 6 to 18, with the numerators unchanged. The same normalization error occurs in the corresponding formulas in [2]. The corrections below are independently checked proposals for discussion, rather than a journal-issued erratum.

Group	Highest weight	Dimension	Coordinate comparison
F4	ω_4	26	Agreement
E6	ω_1	27	Printed values are 3 times too large
E7	ω_7	56	Agreement

1 Scope and conventions

Take the simply connected complex simple group in each case and use Bourbaki numbering. Let $H_i = \alpha_i^\vee$ be the simple coroots and let ω_i be the fundamental weights, so $\omega_i(H_j) = \delta_{ij}$. For $t \in T$, set $t_i = \omega_i(t)$ and write

$$\theta_\lambda(t) = \sum_{i=1}^{\ell} \Theta_i(t) H_i, \Theta_i = \theta_\lambda^*(\omega_i).$$

Thus an exponent vector $a = (a_1, \ldots, a_\ell)$ denotes $t^a = t_1^{a_1} \cdots t_\ell^{a_\ell}$. All displayed coordinate expansions are Laurent polynomials in these variables. They determine the global equivariant morphism, since conjugates of regular torus elements are dense in the group.

2 Trace projection calculation

For $\rho: G \to \mathrm{GL}(V_\lambda)$, project $\rho(g)$ onto $d\rho(\mathfrak{g})$ using the trace form on $End(V_\lambda)$. If $\mu_i = \mu(H_i)$ and m_μ is the multiplicity of the weight μ, the trace equations on the torus give

$$M_{ij} = \sum_\mu m_\mu \mu_i \mu_j, b_j(t) = \sum_\mu m_\mu \mu_j t^\mu, \Theta(t) = M^{-1} b(t).$$

Here M is $S(G, \lambda)$ in [3], and Θ is its coefficient column $c(t)$. The identities $tr(\rho(t) d\rho(H_j)) = b_j(t)$ and $tr(d\rho(H_i) d\rho(H_j)) = M_{ij}$ give this system. The matrix M is nonsingular, so every coordinate is uniquely determined.

3 Independent weights and exact matrices

Starting from the highest weight, apply the simple reflections until its orbit closes. For E6 and E7 the selected representations are minuscule, so every weight in the orbit has multiplicity one. For the 26-dimensional F4 representation, the nonzero weights are the 24 short roots and the zero weight has multiplicity two. These orbit descriptions were separately compared with the full irreducible Weyl characters; the multiplicity dictionaries agree exactly.

Representation	Nonzero orbit size	Zero multiplicity	Laurent terms in all coordinates
F4 ω_4	24	2	70
E6 ω_1	27	0	138
E7 ω_7	56	0	308

The exact inverse Gram matrices are as follows. In each case the matrix inverse was verified by exact rational arithmetic.

$$M_{F_4}^{-1} = \frac{1}{12}\begin{pmatrix} 4 & 6 & 4 & 2 \\ 6 & 12 & 8 & 4 \\ 4 & 8 & 6 & 3 \\ 2 & 4 & 3 & 2 \end{pmatrix}$$

$$M_{E_6}^{-1} = \frac{1}{18}\begin{pmatrix} 4 & 3 & 5 & 6 & 4 & 2 \\ 3 & 6 & 6 & 9 & 6 & 3 \\ 5 & 6 & 10 & 12 & 8 & 4 \\ 6 & 9 & 12 & 18 & 12 & 6 \\ 4 & 6 & 8 & 12 & 10 & 5 \\ 2 & 3 & 4 & 6 & 5 & 4 \end{pmatrix}$$

$$M_{E_7}^{-1} = \frac{1}{24}\begin{pmatrix} 4 & 4 & 6 & 8 & 6 & 4 & 2 \\ 4 & 7 & 8 & 12 & 9 & 6 & 3 \\ 6 & 8 & 12 & 16 & 12 & 8 & 4 \\ 8 & 12 & 16 & 24 & 18 & 12 & 6 \\ 6 & 9 & 12 & 18 & 15 & 10 & 5 \\ 4 & 6 & 8 & 12 & 10 & 8 & 4 \\ 2 & 3 & 4 & 6 & 5 & 4 & 3 \end{pmatrix}$$

Writing $t_j = e^{s_j}$, the same weight sums give two immediate checks:

$$\Theta_i(1) = 0, \frac{\partial \Theta_i(e^s)}{\partial s_j}|_{s=0} = \sum_k (M^{-1})_{ik} M_{kj} = \delta_{ij}.$$

Both identities hold exactly for all 17 coordinates displayed below. The six E6 expressions printed in [1] instead give the matrix $3I_6$ for this derivative. This identifies the common factor of three directly from the normalization of trace projection.

4 Explicit F4 formulas

For $V_{F_4}(\omega_4)$, divide the right side of each displayed identity by 12 to obtain $\Theta_i(t)$. These are identical to the coordinate formulas in [1, Lemma 6.1, pp. 17–18].

Coordinate 1

$$12\Theta_1(t) = 2t_4 + 2t_3t_4^{-1} + 2t_2t_3^{-1} + 2t_1t_2^{-1}t_3 + 2t_1t_3^{-1}t_4 + 2t_1t_4^{-1} - 2t_1^{-1}t_4 - 2t_1^{-1}t_3t_4^{-1} - 2t_1^{-1}t_2t_3^{-1} - 2t_2^{-1}t_3 - 2t_3^{-1}t_4 - 2t_4^{-1}$$

Coordinate 2

$$12\Theta_2(t) = 4t_4 + 4t_3t_4^{-1} + 4t_2t_3^{-1} + 2t_1t_2^{-1}t_3 + 2t_1^{-1}t_3 + 2t_1t_3^{-1}t_4 + 2t_1^{-1}t_2t_3^{-1}t_4 + 2t_1t_4^{-1} + 2t_1^{-1}t_2t_4^{-1} - 2t_1t_2^{-1}t_4 - 2t_1t_2^{-1}t_3t_4^{-1} - 2t_1^{-1}t_4 - 2t_1^{-1}t_3t_4^{-1} - 2t_1t_3^{-1} - 2t_1^{-1}t_2t_3^{-1} - 4t_2^{-1}t_3 - 4t_3^{-1}t_4 - 4t_4^{-1}$$

Coordinate 3

$$12\Theta_3(t) = 3t_4 + 3t_3t_4^{-1} + 2t_2t_3^{-1} + 2t_1t_2^{-1}t_3 + 2t_1^{-1}t_3 + t_1t_3^{-1}t_4 + t_1^{-1}t_2t_3^{-1}t_4 + t_1t_4^{-1} + t_2^{-1}t_3t_4 + t_1^{-1}t_2t_4^{-1} + t_2^{-1}t_3^2t_4^{-1} - t_2t_3^{-2}t_4 - t_2t_3^{-1}t_4^{-1} - t_1t_2^{-1}t_4 - t_1t_2^{-1}t_3t_4^{-1} - t_1^{-1}t_4 - t_1^{-1}t_3t_4^{-1} - 2t_1t_3^{-1} - 2t_1^{-1}t_2t_3^{-1} - 2t_2^{-1}t_3 - 3t_3^{-1}t_4 - 3t_4^{-1}$$

Coordinate 4

$$12\Theta_4(t) = 2t_4 + t_3t_4^{-1} + t_2t_3^{-1} + t_1t_2^{-1}t_3 + t_1^{-1}t_3 + t_1t_3^{-1}t_4 + t_1^{-1}t_2t_3^{-1}t_4 + t_2^{-1}t_3t_4 + t_3^{-1}t_4^2 - t_3t_4^{-2} - t_2t_3^{-1}t_4^{-1} - t_1t_2^{-1}t_3t_4^{-1} - t_1^{-1}t_3t_4^{-1} - t_1t_3^{-1} - t_1^{-1}t_2t_3^{-1} - t_2^{-1}t_3 - t_3^{-1}t_4 - 2t_4^{-1}$$

5 Corrected E6 formulas

For $V_{E_6}(\omega_1)$, divide each right side by 18. The numerators are the printed numerators; the common denominator 6 in [1, Lemma 7.1, pp. 19–20] must be 18. The same repair applies to [2, printed pp. 58–60, PDF pp. 66–68].

Coordinate 1

$$\begin{aligned}18\Theta_1(t) \quad &= 4t_1 + t_1^{-1}t_3 + t_3^{-1}t_4 + t_2t_4^{-1}t_5 + t_2^{-1}t_5 + t_2t_5^{-1}t_6 + t_2^{-1}t_4t_5^{-1}t_6 + t_2t_6^{-1}\\ &+t_3t_4^{-1}t_6 + t_2^{-1}t_4t_6^{-1} + t_1t_3^{-1}t_6 + t_3t_4^{-1}t_5t_6^{-1} - 2t_1^{-1}t_6 + t_1t_3^{-1}t_5t_6^{-1} + t_3t_5^{-1}\\ &-2t_1^{-1}t_5t_6^{-1} + t_1t_3^{-1}t_4t_5^{-1} - 2t_1^{-1}t_4t_5^{-1} + t_1t_2t_4^{-1} - 2t_1^{-1}t_2t_3t_4^{-1} + t_1t_2^{-1}\\ &-2t_1^{-1}t_2^{-1}t_3 - 2t_2t_3^{-1} - 2t_2^{-1}t_3^{-1}t_4 - 2t_4^{-1}t_5 - 2t_5^{-1}t_6 - 2t_6^{-1}\end{aligned}$$

Coordinate 2

$$\begin{aligned}18\Theta_2(t) \quad &= 3t_1 + 3t_1^{-1}t_3 + 3t_3^{-1}t_4 + 3t_2t_4^{-1}t_5 + 3t_2t_5^{-1}t_6 + 3t_2t_6^{-1} - 3t_1t_2^{-1} - 3t_1^{-1}t_2^{-1}t_3\\ &-3t_2^{-1}t_3^{-1}t_4 - 3t_4^{-1}t_5 - 3t_5^{-1}t_6 - 3t_6^{-1}\end{aligned}$$

Coordinate 3

$$\begin{aligned}18\Theta_3(t) \quad &= 5t_1 + 5t_1^{-1}t_3 + 2t_3^{-1}t_4 + 2t_2t_4^{-1}t_5 + 2t_2^{-1}t_5 + 2t_2t_5^{-1}t_6 + 2t_2^{-1}t_4t_5^{-1}t_6 + 2t_2t_6^{-1}\\ &+2t_3t_4^{-1}t_6 + 2t_2^{-1}t_4t_6^{-1} - t_1t_3^{-1}t_6 + 2t_3t_4^{-1}t_5t_6^{-1} - t_1^{-1}t_6 - t_1t_3^{-1}t_5t_6^{-1} + 2t_3t_5^{-1}\\ &-t_1^{-1}t_5t_6^{-1} - t_1t_3^{-1}t_4t_5^{-1} - t_1^{-1}t_4t_5^{-1} - t_1t_2t_4^{-1} - t_1^{-1}t_2t_3t_4^{-1} - t_1t_2^{-1}\\ &-t_1^{-1}t_2^{-1}t_3 - 4t_2t_3^{-1} - 4t_2^{-1}t_3^{-1}t_4 - 4t_4^{-1}t_5 - 4t_5^{-1}t_6 - 4t_6^{-1}\end{aligned}$$

Coordinate 4

$$\begin{aligned}18\Theta_4(t) \quad &= 6t_1 + 6t_1^{-1}t_3 + 6t_3^{-1}t_4 + 3t_2t_4^{-1}t_5 + 3t_2^{-1}t_5 + 3t_2t_5^{-1}t_6 + 3t_2^{-1}t_4t_5^{-1}t_6 + 3t_2t_6^{-1}\\ &+3t_2^{-1}t_4t_6^{-1} - 3t_1t_2t_4^{-1} - 3t_1^{-1}t_2t_3t_4^{-1} - 3t_1t_2^{-1} - 3t_1^{-1}t_2^{-1}t_3 - 3t_2t_3^{-1} - 3t_2^{-1}t_3^{-1}t_4\\ &-6t_4^{-1}t_5 - 6t_5^{-1}t_6 - 6t_6^{-1}\end{aligned}$$

Coordinate 5

$$\begin{aligned}18\Theta_5(t) \quad &= 4t_1 + 4t_1^{-1}t_3 + 4t_3^{-1}t_4 + 4t_2t_4^{-1}t_5 + 4t_2^{-1}t_5 + t_2t_5^{-1}t_6 + t_2^{-1}t_4t_5^{-1}t_6 + t_2t_6^{-1}\\ &+t_3t_4^{-1}t_6 + t_2^{-1}t_4t_6^{-1} + t_1t_3^{-1}t_6 + t_3t_4^{-1}t_5t_6^{-1} + t_1^{-1}t_6 + t_1t_3^{-1}t_5t_6^{-1} - 2t_3t_5^{-1}\\ &+t_1^{-1}t_5t_6^{-1} - 2t_1t_3^{-1}t_4t_5^{-1} - 2t_1^{-1}t_4t_5^{-1} - 2t_1t_2t_4^{-1} - 2t_1^{-1}t_2t_3t_4^{-1} - 2t_1t_2^{-1}\\ &-2t_1^{-1}t_2^{-1}t_3 - 2t_2t_3^{-1} - 2t_2^{-1}t_3^{-1}t_4 - 2t_4^{-1}t_5 - 5t_5^{-1}t_6 - 5t_6^{-1}\end{aligned}$$

Coordinate 6

$$\begin{aligned}18\Theta_6(t) \quad &= 2t_1 + 2t_1^{-1}t_3 + 2t_3^{-1}t_4 + 2t_2t_4^{-1}t_5 + 2t_2^{-1}t_5 + 2t_2t_5^{-1}t_6 + 2t_2^{-1}t_4t_5^{-1}t_6 - t_2t_6^{-1}\\ &+2t_3t_4^{-1}t_6 - t_2^{-1}t_4t_6^{-1} + 2t_1t_3^{-1}t_6 - t_3t_4^{-1}t_5t_6^{-1} + 2t_1^{-1}t_6 - t_1t_3^{-1}t_5t_6^{-1} - t_3t_5^{-1}\\ &-t_1^{-1}t_5t_6^{-1} - t_1t_3^{-1}t_4t_5^{-1} - t_1^{-1}t_4t_5^{-1} - t_1t_2t_4^{-1} - t_1^{-1}t_2t_3t_4^{-1} - t_1t_2^{-1}\\ &-t_1^{-1}t_2^{-1}t_3 - t_2t_3^{-1} - t_2^{-1}t_3^{-1}t_4 - t_4^{-1}t_5 - t_5^{-1}t_6 - 4t_6^{-1}\end{aligned}$$

If the printed coordinates are denoted by $\tilde{\Theta}_i$, then $\tilde{\Theta}_i = 3\Theta_i$ for every i. The resulting polynomial subalgebra over the complex numbers is unchanged, but the displayed functions must be normalized before using the stated Schubert images. Keeping the printed tripled coordinates would instead multiply a homogeneous degree d image by 3^d.

6 Explicit E7 formulas

For $V_{E_7}(\omega_7)$, divide each right side by 24. All seven coordinates agree with [1, Lemma 8.1, pp. 24–25]. No normalization change is needed.

Coordinate 1

$$\begin{aligned}24\Theta_1(t) \;=\; & 2t_7 + 2t_6t_7^{-1} + 2t_5t_6^{-1} + 2t_4t_5^{-1} + 2t_2t_3t_4^{-1} + 2t_2^{-1}t_3 + 2t_1t_2t_3^{-1} + 2t_1t_2^{-1}t_3^{-1}t_4 \\ & + 2t_1t_4^{-1}t_5 + 2t_1t_5^{-1}t_6 + 2t_1t_6^{-1}t_7 + 2t_1t_7^{-1} - 2t_1^{-1}t_7 - 2t_1^{-1}t_6t_7^{-1} - 2t_1^{-1}t_5t_6^{-1} \\ & - 2t_1^{-1}t_4t_5^{-1} - 2t_1^{-1}t_2t_3t_4^{-1} - 2t_1^{-1}t_2^{-1}t_3 - 2t_2t_3^{-1} - 2t_2^{-1}t_3^{-1}t_4 - 2t_4^{-1}t_5 - 2t_5^{-1}t_6 \\ & - 2t_6^{-1}t_7 - 2t_7^{-1}\end{aligned}$$

Coordinate 2

$$\begin{aligned}24\Theta_2(t) \;=\; & 3t_7 + 3t_6t_7^{-1} + 3t_5t_6^{-1} + 3t_4t_5^{-1} + 3t_2t_3t_4^{-1} + t_2^{-1}t_3 + 3t_1t_2t_3^{-1} + t_1t_2^{-1}t_3^{-1}t_4 \\ & + 3t_1^{-1}t_2 + t_1^{-1}t_2^{-1}t_4 + t_1t_4^{-1}t_5 + t_1^{-1}t_3t_4^{-1}t_5 + t_1t_5^{-1}t_6 + t_3^{-1}t_5 + t_1^{-1}t_3t_5^{-1}t_6 \\ & + t_1t_6^{-1}t_7 + t_3^{-1}t_4t_5^{-1}t_6 + t_1^{-1}t_3t_6^{-1}t_7 + t_1t_7^{-1} + t_2t_4^{-1}t_6 + t_3^{-1}t_4t_6^{-1}t_7 \\ & + t_1^{-1}t_3t_7^{-1} - t_2^{-1}t_6 + t_2t_4^{-1}t_5t_6^{-1}t_7 + t_3^{-1}t_4t_7^{-1} - t_2^{-1}t_5t_6^{-1}t_7 + t_2t_5^{-1}t_7 \\ & + t_2t_4^{-1}t_5t_7^{-1} - t_2^{-1}t_4t_5^{-1}t_7 - t_2^{-1}t_5t_7^{-1} + t_2t_5^{-1}t_6t_7^{-1} - t_3t_4^{-1}t_7 \\ & - t_2^{-1}t_4t_5^{-1}t_6t_7^{-1} + t_2t_6^{-1} - t_1t_3^{-1}t_7 - t_3t_4^{-1}t_6t_7^{-1} - t_2^{-1}t_4t_6^{-1} - t_1^{-1}t_7 \\ & - t_1t_3^{-1}t_6t_7^{-1} - t_3t_4^{-1}t_5t_6^{-1} - t_1^{-1}t_6t_7^{-1} - t_1t_3^{-1}t_5t_6^{-1} - t_3t_5^{-1} - t_1^{-1}t_5t_6^{-1} \\ & - t_1t_3^{-1}t_4t_5^{-1} - t_1^{-1}t_4t_5^{-1} - t_1t_2t_4^{-1} - t_1^{-1}t_2t_3t_4^{-1} - 3t_1t_2^{-1} - 3t_1^{-1}t_2^{-1}t_3 - t_2t_3^{-1} \\ & - 3t_2^{-1}t_3^{-1}t_4 - 3t_4^{-1}t_5 - 3t_5^{-1}t_6 - 3t_6^{-1}t_7 - 3t_7^{-1}\end{aligned}$$

Coordinate 3

$$\begin{aligned}24\Theta_3(t) \;=\; & 4t_7 + 4t_6t_7^{-1} + 4t_5t_6^{-1} + 4t_4t_5^{-1} + 4t_2t_3t_4^{-1} + 4t_2^{-1}t_3 + 2t_1t_2t_3^{-1} + 2t_1t_2^{-1}t_3^{-1}t_4 \\ & + 2t_1^{-1}t_2 + 2t_1^{-1}t_2^{-1}t_4 + 2t_1t_4^{-1}t_5 + 2t_1^{-1}t_3t_4^{-1}t_5 + 2t_1t_5^{-1}t_6 + 2t_1^{-1}t_3t_5^{-1}t_6 \\ & + 2t_1t_6^{-1}t_7 + 2t_1^{-1}t_3t_6^{-1}t_7 + 2t_1t_7^{-1} + 2t_1^{-1}t_3t_7^{-1} - 2t_1t_3^{-1}t_7 - 2t_1^{-1}t_7 - 2t_1t_3^{-1}t_6t_7^{-1} \\ & - 2t_1^{-1}t_6t_7^{-1} - 2t_1t_3^{-1}t_5t_6^{-1} - 2t_1^{-1}t_5t_6^{-1} - 2t_1t_3^{-1}t_4t_5^{-1} - 2t_1^{-1}t_4t_5^{-1} - 2t_1t_2t_4^{-1} \\ & - 2t_1^{-1}t_2t_3t_4^{-1} - 2t_1t_2^{-1} - 2t_1^{-1}t_2^{-1}t_3 - 4t_2t_3^{-1} - 4t_2^{-1}t_3^{-1}t_4 - 4t_4^{-1}t_5 - 4t_5^{-1}t_6 \\ & - 4t_6^{-1}t_7 - 4t_7^{-1}\end{aligned}$$

Coordinate 4

$$\begin{aligned}24\Theta_4(t) \;=\; & 6t_7 + 6t_6t_7^{-1} + 6t_5t_6^{-1} + 6t_4t_5^{-1} + 4t_2t_3t_4^{-1} + 4t_2^{-1}t_3 + 4t_1t_2t_3^{-1} + 4t_1t_2^{-1}t_3^{-1}t_4 \\ & + 4t_1^{-1}t_2 + 4t_1^{-1}t_2^{-1}t_4 + 2t_1t_4^{-1}t_5 + 2t_1^{-1}t_3t_4^{-1}t_5 + 2t_1t_5^{-1}t_6 + 2t_3^{-1}t_5 + 2t_1^{-1}t_3t_5^{-1}t_6 \\ & + 2t_1t_6^{-1}t_7 + 2t_3^{-1}t_4t_5^{-1}t_6 + 2t_1^{-1}t_3t_6^{-1}t_7 + 2t_1t_7^{-1} + 2t_3^{-1}t_4t_6^{-1}t_7 + 2t_1^{-1}t_3t_7^{-1} \\ & + 2t_3^{-1}t_4t_7^{-1} - 2t_3t_4^{-1}t_7 - 2t_1t_3^{-1}t_7 - 2t_3t_4^{-1}t_6t_7^{-1} - 2t_1^{-1}t_7 - 2t_1t_3^{-1}t_6t_7^{-1} \\ & - 2t_3t_4^{-1}t_5t_6^{-1} - 2t_1^{-1}t_6t_7^{-1} - 2t_1t_3^{-1}t_5t_6^{-1} - 2t_3t_5^{-1} - 2t_1^{-1}t_5t_6^{-1} - 2t_1t_3^{-1}t_4t_5^{-1} \\ & - 2t_1^{-1}t_4t_5^{-1} - 4t_1t_2t_4^{-1} - 4t_1^{-1}t_2t_3t_4^{-1} - 4t_1t_2^{-1} - 4t_1^{-1}t_2^{-1}t_3 - 4t_2t_3^{-1} - 4t_2^{-1}t_3^{-1}t_4 \\ & - 6t_4^{-1}t_5 - 6t_5^{-1}t_6 - 6t_6^{-1}t_7 - 6t_7^{-1}\end{aligned}$$

E7 formulas continued

Continue to divide each right side by 24.

Coordinate 5

$$\begin{aligned}24\Theta_5(t) \quad &= 5t_7 + 5t_6t_7^{-1} + 5t_5t_6^{-1} + 3t_4t_5^{-1} + 3t_2t_3t_4^{-1} + 3t_2^{-1}t_3 + 3t_1t_2t_3^{-1} + 3t_1t_2^{-1}t_3^{-1}t_4 \\ &+3t_1^{-1}t_2 + 3t_1^{-1}t_2^{-1}t_4 + 3t_1t_4^{-1}t_5 + 3t_1^{-1}t_3t_4^{-1}t_5 + t_1t_5^{-1}t_6 + 3t_3^{-1}t_5 + t_1^{-1}t_3t_5^{-1}t_6 \\ &+t_1t_6^{-1}t_7 + t_3^{-1}t_4t_5^{-1}t_6 + t_1^{-1}t_3t_6^{-1}t_7 + t_1t_7^{-1} + t_2t_4^{-1}t_6 + t_3^{-1}t_4t_6^{-1}t_7 \\ &+t_1^{-1}t_3t_7^{-1} + t_2^{-1}t_6 + t_2t_4^{-1}t_5t_6^{-1}t_7 + t_3^{-1}t_4t_7^{-1} + t_2^{-1}t_5t_6^{-1}t_7 - t_2t_5^{-1}t_7 \\ &+t_2t_4^{-1}t_5t_7^{-1} - t_2^{-1}t_4t_5^{-1}t_7 + t_2^{-1}t_5t_7^{-1} - t_2t_5^{-1}t_6t_7^{-1} - t_3t_4^{-1}t_7 \\ &-t_2^{-1}t_4t_5^{-1}t_6t_7^{-1} - t_2t_6^{-1} - t_1t_3^{-1}t_7 - t_3t_4^{-1}t_6t_7^{-1} - t_2^{-1}t_4t_6^{-1} - t_1^{-1}t_7 \\ &-t_1t_3^{-1}t_6t_7^{-1} - t_3t_4^{-1}t_5t_6^{-1} - t_1^{-1}t_6t_7^{-1} - t_1t_3^{-1}t_5t_6^{-1} - 3t_3t_5^{-1} - t_1^{-1}t_5t_6^{-1} \\ &-3t_1t_3^{-1}t_4t_5^{-1} - 3t_1^{-1}t_4t_5^{-1} - 3t_1t_2t_4^{-1} - 3t_1^{-1}t_2t_3t_4^{-1} - 3t_1t_2^{-1} - 3t_1^{-1}t_2^{-1}t_3 - 3t_2t_3^{-1} \\ &-3t_2^{-1}t_3^{-1}t_4 - 3t_4^{-1}t_5 - 5t_5^{-1}t_6 - 5t_6^{-1}t_7 - 5t_7^{-1}\end{aligned}$$

Coordinate 6

$$\begin{aligned}24\Theta_6(t) \quad &= 4t_7 + 4t_6t_7^{-1} + 2t_5t_6^{-1} + 2t_4t_5^{-1} + 2t_2t_3t_4^{-1} + 2t_2^{-1}t_3 + 2t_1t_2t_3^{-1} + 2t_1t_2^{-1}t_3^{-1}t_4 \\ &+2t_1^{-1}t_2 + 2t_1^{-1}t_2^{-1}t_4 + 2t_1t_4^{-1}t_5 + 2t_1^{-1}t_3t_4^{-1}t_5 + 2t_1t_5^{-1}t_6 + 2t_3^{-1}t_5 + 2t_1^{-1}t_3t_5^{-1}t_6 \\ &+2t_3^{-1}t_4t_5^{-1}t_6 + 2t_2t_4^{-1}t_6 + 2t_2^{-1}t_6 - 2t_2t_6^{-1} - 2t_2^{-1}t_4t_6^{-1} - 2t_3t_4^{-1}t_5t_6^{-1} \\ &-2t_1t_3^{-1}t_5t_6^{-1} - 2t_3t_5^{-1} - 2t_1^{-1}t_5t_6^{-1} - 2t_1t_3^{-1}t_4t_5^{-1} - 2t_1^{-1}t_4t_5^{-1} - 2t_1t_2t_4^{-1} \\ &-2t_1^{-1}t_2t_3t_4^{-1} - 2t_1t_2^{-1} - 2t_1^{-1}t_2^{-1}t_3 - 2t_2t_3^{-1} - 2t_2^{-1}t_3^{-1}t_4 - 2t_4^{-1}t_5 - 2t_5^{-1}t_6 \\ &-4t_6^{-1}t_7 - 4t_7^{-1}\end{aligned}$$

Coordinate 7

$$\begin{aligned}24\Theta_7(t) \quad &= 3t_7 + t_6t_7^{-1} + t_5t_6^{-1} + t_4t_5^{-1} + t_2t_3t_4^{-1} + t_2^{-1}t_3 + t_1t_2t_3^{-1} + t_1t_2^{-1}t_3^{-1}t_4 \\ &+t_1^{-1}t_2 + t_1^{-1}t_2^{-1}t_4 + t_1t_4^{-1}t_5 + t_1^{-1}t_3t_4^{-1}t_5 + t_1t_5^{-1}t_6 + t_3^{-1}t_5 + t_1^{-1}t_3t_5^{-1}t_6 \\ &+t_1t_6^{-1}t_7 + t_3^{-1}t_4t_5^{-1}t_6 + t_1^{-1}t_3t_6^{-1}t_7 - t_1t_7^{-1} + t_2t_4^{-1}t_6 + t_3^{-1}t_4t_6^{-1}t_7 \\ &-t_1^{-1}t_3t_7^{-1} + t_2^{-1}t_6 + t_2t_4^{-1}t_5t_6^{-1}t_7 - t_3^{-1}t_4t_7^{-1} + t_2^{-1}t_5t_6^{-1}t_7 + t_2t_5^{-1}t_7 \\ &-t_2t_4^{-1}t_5t_7^{-1} + t_2^{-1}t_4t_5^{-1}t_7 - t_2^{-1}t_5t_7^{-1} - t_2t_5^{-1}t_6t_7^{-1} + t_3t_4^{-1}t_7 \\ &-t_2^{-1}t_4t_5^{-1}t_6t_7^{-1} - t_2t_6^{-1} + t_1t_3^{-1}t_7 - t_3t_4^{-1}t_6t_7^{-1} - t_2^{-1}t_4t_6^{-1} + t_1^{-1}t_7 \\ &-t_1t_3^{-1}t_6t_7^{-1} - t_3t_4^{-1}t_5t_6^{-1} - t_1^{-1}t_6t_7^{-1} - t_1t_3^{-1}t_5t_6^{-1} - t_3t_5^{-1} - t_1^{-1}t_5t_6^{-1} \\ &-t_1t_3^{-1}t_4t_5^{-1} - t_1^{-1}t_4t_5^{-1} - t_1t_2t_4^{-1} - t_1^{-1}t_2t_3t_4^{-1} - t_1t_2^{-1} - t_1^{-1}t_2^{-1}t_3 - t_2t_3^{-1} \\ &-t_2^{-1}t_3^{-1}t_4 - t_4^{-1}t_5 - t_5^{-1}t_6 - t_6^{-1}t_7 - 3t_7^{-1}\end{aligned}$$

The comparison includes every printed monomial and any monomial that could have been omitted. All supports coincide. The only coordinate discrepancy among the three requested representations is the common E6 scalar factor described above. The Weyl-equivariance of these formulas follows from the Weyl invariance of the weight multiplicities and the trace form.

Appendix A Corrections to Schubert expansions

This appendix records further discrepancies in the later Schubert tables, separately from the Springer-coordinate calculation. It uses the $y_j^{L_r}$ and Schubert conventions of [1]. All E6 expressions here use the normalized coordinates with denominator 18.

Location in [1]	Coefficient in print	Correct coefficient
Theorem 6.2b F4 P2 e3 page 18	−36	44
Theorem 7.2c E6 P3 e4 page 22	−31	31
Theorem 7.2f E6 P6 squared e1 page 23	−5	−3
Theorem 7.2f E6 P6 squared e2 page 23	4254	42

$$\xi^{P_2}(e_3(y_3^{L_2}, y_4^{L_2}, y_5^{L_2})) = \quad -4\epsilon^{P_2}_{s_3s_1s_2} - 10\epsilon^{P_2}_{s_2s_3s_2} + 44\epsilon^{P_2}_{s_4s_3s_2}$$

$$\xi^{P_3}(e_4(y_3^{L_3}, \dots, y_7^{L_3})) = \quad 31\epsilon^{P_3}_{s_2s_4s_1s_3} - 38\epsilon^{P_3}_{s_3s_4s_1s_3} + 31\epsilon^{P_3}_{s_5s_4s_1s_3} - 118\epsilon^{P_3}_{s_5s_2s_4s_3} + 325\epsilon^{P_3}_{s_6s_5s_4s_3}$$

$$\xi^{P_6}(e_1((y_2^{L_6})^2, \dots, (y_6^{L_6})^2)) = \quad -3\epsilon^{P_6}_{s_5s_6}$$

$$\xi^{P_6}(e_2((y_2^{L_6})^2, \dots, (y_6^{L_6})^2)) = \quad -54\epsilon^{P_6}_{s_2s_4s_5s_6} + 42\epsilon^{P_6}_{s_3s_4s_5s_6}$$

Each complete corrected row was independently evaluated by exact BGG divided differences and by generalized Chevalley multiplication over the positive roots. Both algorithms agree. The complete published Part III coefficient audit covers 118 rows and 335 coefficients, including the E7 degree 12 invariant.

Two additional E6 repairs occur in the same theorem. On p. 22, in the P3 e5 row, insert a plus sign before $492\epsilon^{P_3}_{s_6s_5s_4s_1s_3}$. On p. 23, the P6 ring description includes the Euler-product generator but its image is absent from the generator list. Its image, verified by both algorithms, is

$$\xi^{P_6}(y_2^{L_6}y_3^{L_6}y_4^{L_6}y_5^{L_6}y_6^{L_6}) = 6\epsilon^{P_6}_{s_3s_2s_4s_5s_6} - 21\epsilon^{P_6}_{s_1s_3s_4s_5s_6}.$$

These findings do not identify the historical cause of any discrepancy or establish which errors had previously been noticed. They concern explicit formulas and do not supply a counterexample to the classical stable-injectivity results.

Appendix B Reproducible exact calculation

The following standalone script was executed under SageMath 9.5. It reconstructs the weights by reflections, checks their multiplicities against the irreducible character, prints all Gram matrices and coordinates, and checks the value and derivative at the identity. It reads no published coefficients or external data files. Save the script as reproduce_springer.py and run sage -python reproduce_springer.py.

```
"""Run with: sage -python reproduce_springer.py (tested with SageMath 9.5).

Bourbaki numbering; H_i = alpha_i^vee and t_i = omega_i(t).
The Springer coordinates satisfy theta(t) = sum_i Theta_i(t) H_i.
No published formula or previous computation is imported.
"""
from sage.all import (QQ, RootSystem, WeylCharacterRing, matrix, vector,
                      LaurentPolynomialRing, identity_matrix, prod)

cases = [('F4', (0, 0, 0, 1)),
         ('E6', (1, 0, 0, 0, 0, 0)),
         ('E7', (0, 0, 0, 0, 0, 0, 1))]

for kind, highest in cases:
    r = len(highest)
    ambient = RootSystem(kind).ambient_space()
    roots = ambient.simple_roots()
    coroots = ambient.simple_coroots()
    alpha = [tuple(roots[i+1].inner_product(coroots[j+1])
                   for j in range(r)) for i in range(r)]

    # Reflect a weight mu by s_i(mu) = mu - mu(H_i) alpha_i.
    orbit, pending = {highest}, [highest]
    while pending:
        mu = pending.pop()
        for i in range(r):
            nu = tuple(mu[j] - mu[i]*alpha[i][j] for j in range(r))
            if nu not in orbit:
                orbit.add(nu)
                pending.append(nu)
    multiplicities = {mu: 1 for mu in orbit}
    if kind == 'F4':
        multiplicities[(0,)*r] = 2

    # Cross-check the orbit construction against the irreducible character.
    character = WeylCharacterRing(kind, style='coroots')(*highest)
```

Reproducible calculation continued

```
check = {tuple(w.inner_product(coroots[i+1]) for i in range(r)): int(m)
         for w, m in character.weight_multiplicities().items()}
assert check == multiplicities

M = matrix(QQ, r, r, lambda i, j: sum(
    m*mu[i]*mu[j] for mu, m in multiplicities.items()))
L = LaurentPolynomialRing(QQ, names=['t%d' % (i+1) for i in range(r)])
t = L.gens()
monomial = {mu: prod(t[i]**int(mu[i]) for i in range(r))
            for mu in multiplicities}
b = vector(L, [sum(m*mu[j]*monomial[mu]
                   for mu, m in multiplicities.items()) for j in range(r)])
Theta = M.inverse()*b
D = matrix(QQ, r, r, lambda i, j:
           Theta[i].derivative(t[j])(*([1]*r)))
assert D == identity_matrix(QQ, r)
assert all(f(*([1]*r)) == 0 for f in Theta)

print('\n%s: dimension %s, nonzero orbit size %s' %
      (kind, sum(multiplicities.values()), len(orbit)))
print('M =\n%s\nM^-1 =\n%s' % (M, M.inverse()))
for i, f in enumerate(Theta, 1):
    print('Theta_%d = %s' % (i, f))
print('d(theta)_1 = I: verified exactly')
```

References

[1] Shrawan Kumar and Jiale Xie. Representation ring of Levi subgroups versus cohomology ring of flag varieties III. Journal of Pure and Applied Algebra 228 (2024), 107539. DOI 10.1016/j.jpaa.2023.107539. The journal PDF is available at https://kumar.math.unc.edu/papers/kumar86.pdf. Page references above use this 30-page journal PDF.

[2] Jiale Xie. Polynomial Representation Ring versus Cohomology Ring of Flag Varieties for Classical and Exceptional Groups. Ph.D. dissertation, University of North Carolina at Chapel Hill, 2022. The inspected PDF has 109 pages; the E6 coordinates occupy printed pages 58–60, PDF pages 66–68.

[3] Sean Rogers. An explicit determination of the Springer morphism. Communications in Algebra 46 (2018), 4233–4242. DOI 10.1080/00927872.2017.1407418. Preprint: https://arxiv.org/abs/1701.01538. Theorems 1 and 2 supply the uniform weight-matrix method applied here; [1] explicitly cites these theorems for its exceptional Springer calculations.

Reproducibility record. All arithmetic uses rational coefficients. The independent reconstruction checks 17 coordinate polynomials and all 516 printed Laurent monomials for these three representations. The E6 printed-to-normalized ratio is exactly 3 for every nonzero coefficient; the F4 and E7 ratios are exactly 1. This note verifies the numerical evaluations in [1] and [2]; it does not reaudit the proofs of the general theorems in [3].